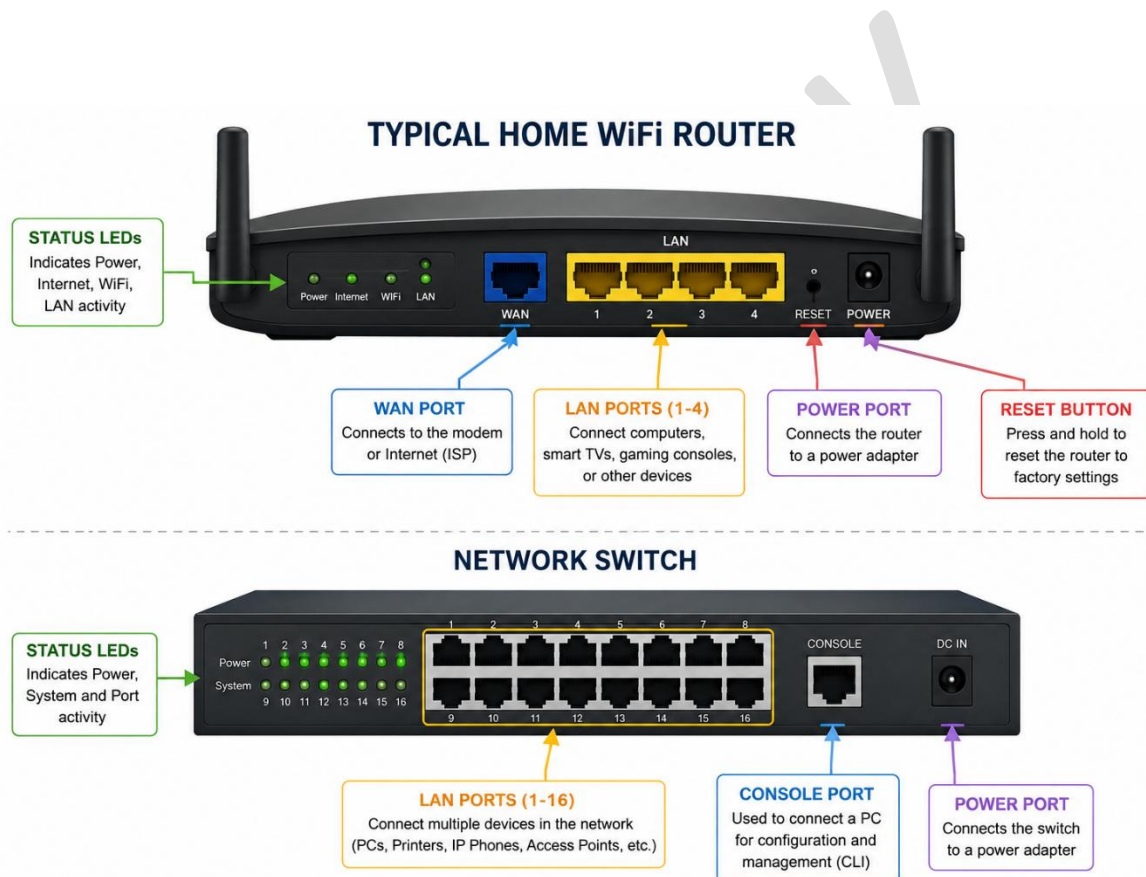


A+: UNDERSTANDING OF NETWORK

TASK: 1

Draw a labelled diagram of a typical home WiFi router and a network switch, clearly marking at least 3 different ports (such as WAN, LAN, Console) and the location of status LEDs.



TASK: 2

List 3 key differences between routers and switches in terms of their main functions and how they handle network traffic.

Hint: Think about how each device deals with IP addresses, MAC addresses, and the types of networks they connect.

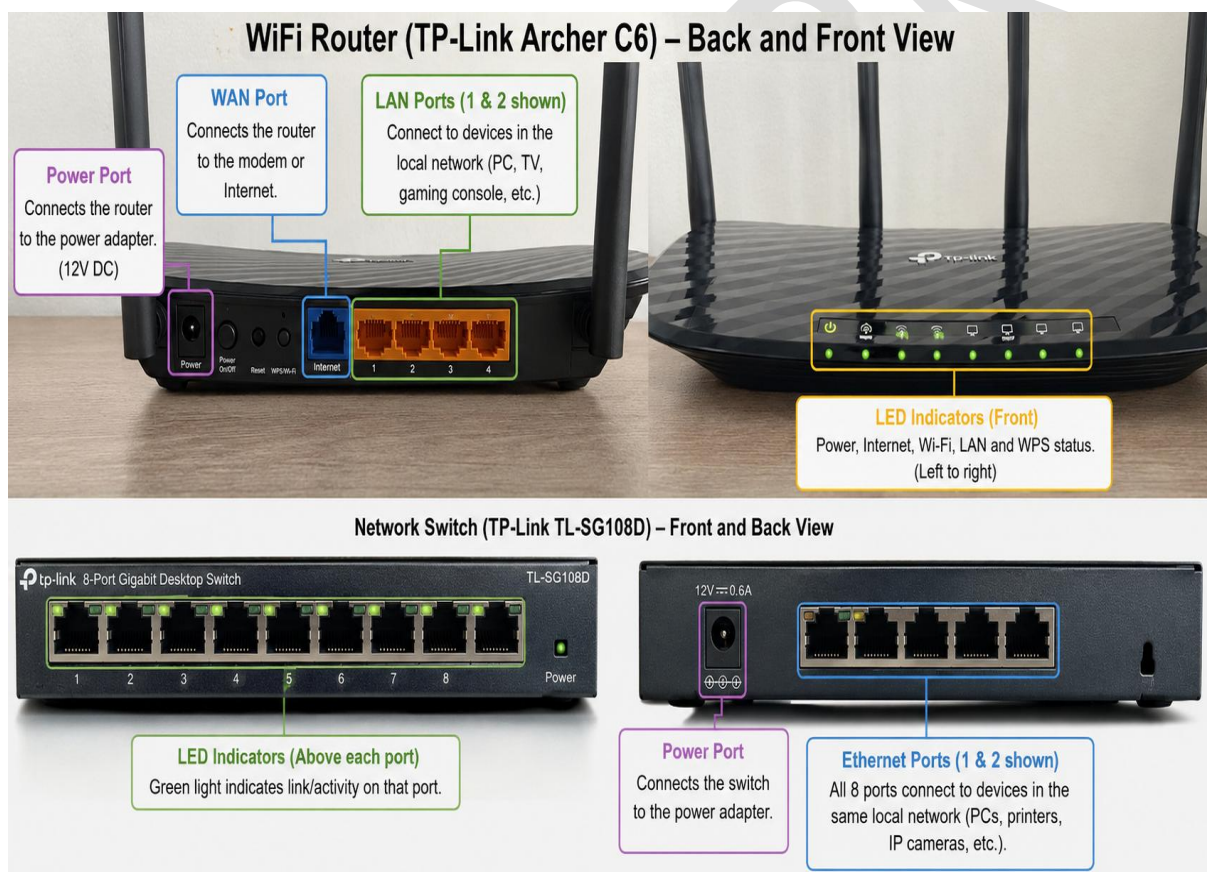
Difference	Router	Switch
Main function	Connects different networks , such as a home network to the Internet.	Connects multiple devices within the same local network (LAN) .
Address used	Uses IP addresses to determine where data should be sent between networks.	Uses MAC addresses to send data to the correct device within a LAN.
Traffic handling	Routes traffic between networks and can choose the best path for data.	Forwards traffic directly to the correct connected port, reducing unnecessary network traffic.

In simple words:

A **router connects networks**, while a **switch connects devices within a network**.

TASK: 3

Take a real or virtual image of a router and a switch (from your home, lab, or Google Images), and identify the following on each: Power port, at least 2 Ethernet ports, and any LED indicators. Annotate the image or describe their position and purpose.



TASK: 4

Imagine you are setting up a small gaming café network for IPL streaming and multiplayer matches. Write a short scenario explaining which device (router or switch) you would connect first to the ISP line, and how you would connect 8 gaming PCs to the network. Justify your choices.

For a small gaming café, I would connect the **ISP Internet line to the router first**. The router would handle the Internet connection, assign IP addresses to devices, and provide security through features such as NAT and a firewall.

Next, I would connect one **LAN port of the router to an 8-port network switch** using an Ethernet cable. Then, I would connect each of the **8 gaming PCs to the switch**, with one Ethernet cable going from each PC to a separate switch port.

Network setup:

ISP Line → Router → 8-Port Switch → 8 Gaming PCs

This setup is best because the **router connects the café network to the Internet**, which is necessary for IPL streaming and online multiplayer games. The **switch provides enough Ethernet ports to connect all 8 PCs** and efficiently handles local network traffic between them. Using wired connections also provides more stable speeds and lower latency than Wi-Fi, which is important for multiplayer gaming.